Keewoo Lee

keewoole@gmail.com https://keewoolee.github.io

OVERVIEW

I am a cryptography researcher at Ethereum Foundation. Previously, I was a postdoctoral researcher at UC Berkeley, hosted by Prof. Sanjam Garg. I obtained my Ph.D. in Mathematical Sciences at Seoul National University, advised by Prof. Jung Hee Cheon. I am broadly interested in cryptography from theory to practice. Currently, my research focus is on cryptographic primitives for secure computation (e.g., homomorphic encryption, secure multiparty computation) and their applications.

EMPLOYMENT

Ethereum Foundation, Remote

- Scientific Advisor (Full-time) — Dec 2025 – Present
  - Access Layer Cluster
- Scientific Advisor (Part-time) — Feb 2025 – Nov 2025
  - Privacy Cluster

University of California, Berkeley, United States

- Postdoctoral Researcher — Nov 2023 – Nov 2025
  - Host: Prof. Sanjam Garg

CryptoLab Inc., Republic of Korea

- Research Scientist (Freelancer), HealthcareAI Division — Sep 2023 – Oct 2023
  - Focus: Privacy-preserving Machine Learning on Biomedical Data

EDUCATION

Seoul National University, Republic of Korea

- Ph.D. in Mathematical Sciences — Sep 2017 – Aug 2023
  - Advisor: Prof. Jung Hee Cheon
  - Focus: Cryptography (Homomorphic Encryption, Secure Multiparty Computation, Lattice-based Cryptography)
  - Thesis: "A Study on Homomorphic Packing: Definitions, Constructions, and Limitations"
- B.S. in Mathematical Sciences — Mar 2014 – Aug 2017

PUBLICATIONS

Authors are listed in alphabetical order by last name, unless an asterisk(*) is indicated.

CONFERENCE & JOURNAL

[C22] Charles Gouert, K. Lee, Dimitris Mouris, Yiannis Tselekounis, Jean-Luc Watson, "StOMR: Stateful Oblivious Message Retrieval," *ACM SIGSAC Conference on Computer and Communications Security (CCS 2026)*

[C21] K. Lee, "Barely Doubly-Efficient SimplePIR," *Annual International Cryptology Conference (Crypto 2026)*

[C20] Jaiden Fairoze, Sanjam Garg, K. Lee, Mingyuan Wang, "Bypassing Prompt Guards in Production with Controlled-Release Prompting," *USENIX Security Symposium (USENIX Security 2026)*

[C19] K. Lee, Yongdong Yeo, "SophOMR: Improved Oblivious Message Retrieval from SIMD-Aware Homomorphic Compression," *USENIX Security Symposium (USENIX Security 2026)*

[C18] Leo de Castro, Duhyeong Kim, Miran Kim, K. Lee, Seonhong Min, Yongsoo Song, "More Efficient Lattice-based OLE from Circuit-private Linear HE with Linear Overhead," *Communications in Cryptology (IACR CiC)*, 2026

[C17] *Mayank Rathee, K. Lee, Raluca Ada Popa, "Verifiable PIR with Small Client Storage," *IEEE Symposium on Security and Privacy (S&P 2026)*

[C16] *Grant Bosworth, K. Lee, Sunwoong Kim, "Leveraging FPGAs for Homomorphic Matrix-Vector Multiplication in Oblivious Message Retrieval," *Asilomar Conference on Signals, Systems, and Computers (ACSSC 2025)*

[C15] Leo de Castro, K. Lee, "VeriSimplePIR: Verifiability in SimplePIR at No Online Cost for Honest Servers," *USENIX Security Symposium (USENIX Security 2024)*

[C14] K. Lee, "Bit Security as Cost to Demonstrate Advantage," *Communications in Cryptology (IACR CiC)*, 2024

- Best Award, National Cryptography Contest 2022

[C13] *Seoyoung Ko, K. Lee, Hyunhum Cho, Yoonjae Hwang, Huisu Jang, "Asynchronous Federated Learning with Directed Acyclic Graph-based Blockchain in Edge Computing: Overview, Design, and Challenges," *Expert Systems with Applications*, 2023

[C12] Jung Hee Cheon, K. Lee, "Limits of Polynomial Packings for $\mathbb{Z}_{p^k}$ and $\mathbb{F}_{p^k}$," *Annual International Conference on the Theory and Applications of Cryptographic Techniques (Eurocrypt 2022)*

- Best Award, National Cryptography Contest 2021

[C11] Jung Hee Cheon, Dongwoo Kim, Duhyeong Kim, K. Lee, "On the Scaled Inverse of $(x^i - x^j)$ modulo Cyclotomic Polynomial of the form $\Phi_{p^s}(x)$ or $\Phi_{p^s q^t}(x)$," *Journal of the Korean Mathematical Society*, 2022

[C10] *Michael Cho, K. Lee, Sunwoong Kim, "HELPSE: Homomorphic Encryption-based Lightweight Password Strength Estimation in a Virtual Keyboard System," *Great Lakes Symposium on VLSI (GLSVLSI 2022)*

[C09] Jung Hee Cheon, Dongwoo Kim, K. Lee, "MHz2k: MPC from HE over $\mathbb{Z}_{2^k}$ with New Packing, Simpler Reshare, and Better ZKP," *Annual International Cryptology Conference (Crypto 2021)*

- Excellence Award, National Cryptography Contest 2020

[C08] *Wonkyung Jung, Eojin Lee, Sangpyo Kim, K. Lee, Namhoon Kim, Chohong Min, Jung Hee Cheon, Jung Ho Ahn, "Accelerating Fully Homomorphic Encryption Through Architecture-Centric Analysis and Optimization," *IEEE Access*, 2021

- A preliminary version appeared in *ISPASS 2021*.

[C07] *Sunwoong Kim, K. Lee, Wonhee Cho, Yujin Nam, Jung Hee Cheon, Rob A. Rutenbar, "Hardware Architecture of a Number Theoretic Transform for a Bootstrappable RNS-based Homomorphic Encryption Scheme," *2020 IEEE 28th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM 2020)*

[C06] Jung Hee Cheon, Dongwoo Kim, Duhyeong Kim, Hun Hee Lee, K. Lee, "Numerical Methods for Comparison on Homomorphically Encrypted Numbers," *International Conference on the Theory and Applications of Cryptology and Information Security (Asiacrypt 2019)*

- Invited to *Journal of Cryptology* (Top 3 of 71 accepted papers among 307 submissions)

[C05] *Sunwoong Kim, K. Lee, Wonhee Cho, Jung Hee Cheon, Rob A. Rutenbar, "FPGA-based Accelerators of Fully Pipelined Modular Multipliers for Homomorphic Encryption," *2019 International Conference on ReConFigurable Computing and FPGAs (ReConFig 2019)*

[C04] *Sungjoon Park, Minsu Kim, Seokjun Seo, Seungwan Hong, Kyoohyung Han, K. Lee, Jung Hee Cheon, Sun Kim, "A Secure SNP Panel Scheme using Homomorphically Encrypted K-mers without SNP Calling on the User Side," *BMC Genomics*, 2019

[C03] Jung Hee Cheon, Haejin Cho, Jaewook Jung, Joohee Lee, K. Lee, "Efficient Identity-Based Encryption from LWR," *Annual International Conference on Information Security and Cryptology (ICISC 2019)*

[C02] *Andrey Kim, Yongsoo Song, Miran Kim, K. Lee, Jung Hee Cheon, "Logistic Regression Model Training based on the Approximate Homomorphic Encryption," *BMC Medical Genomics*, 2018

- First Place Prize, iDASH Genomic Data Privacy and Security Protection Competition 2017

[C01] Jung Hee Cheon, Jinhyuck Jeong, Joohee Lee, K. Lee, "Privacy-preserving Computations of Predictive Medical Models with Minimax Approximation and Non-adjacent Form," *International Conference on Financial Cryptography and Data Security (WAHC 2017)*

- Excellence Award, National Cryptography Contest 2016

BOOK CHAPTERS

[B01] Laia Amorós, Syed Mahbub Hafiz, K. Lee, M. Caner Tol, "Gimme That Model!: A Trusted ML Model Trading Protocol," In *Protecting Privacy through Homomorphic Encryption*, 2021

PREPRINTS

[P09] K. Lee, "Barely Sublinear RAM-FHE from Plain LWE," https://eprint.iacr.org/2026/2412

[P08] K. Lee, Yongha Son, "SkrrtPIR: Doubly-Stateless Batch PIR from Single-Key RLWE Unpacking," https://eprint.iacr.org/2026/2385

[P07] K. Lee, "Pseudorandom Codes from LWE," https://eprint.iacr.org/2026/2339

[P06] K. Lee, "Barely Doubly-Efficient PIR from Quadratic Residuosity," https://eprint.iacr.org/2026/2034

[P05] K. Lee, "Information-theoretic two-server PIR requires $(6(1-o(1))\log n$ bits of communication," https://eprint.iacr.org/2026/1959

[P04] *Seewoo Lee, Byung-Hak Hwang, Hyojae Lim, Jihoon Hyun, Ilkyoo Choi, Yeachan Park, Jineon Baek, Hyukpyo Hong, K. Lee, Jaeseong Heo, Hyungryul Baik, Chul-hee Lee, Kyu-Hwan Lee, "Lean-GAP: A Dataset of Formalized Graduate Algebra Problems," https://arxiv.org/abs/2606.02588

[P03] Arka Rai Choudhuri, Sanjam Garg, K. Lee, Hart Montgomery, Guru Vamsi Policharla, Rohit Sinha, "A Cryptographic Framework for Proof of Personhood," https://eprint.iacr.org/2026/333

[P02] Arka Rai Choudhuri, Sanjam Garg, Matthew Gregoire, K. Lee, Mike Lodder, Hart Montgomery, Guru Vamsi Policharla, Jim Zhang, "Qurrency: a quantum-secure, private, and auditable platform for digital assets," https://eprint.iacr.org/2026/015

[P01] Jung Hee Cheon, K. Lee, Jai Hyun Park, Yongdong Yeo, "Private Database Query with SIMD-Aware Homomorphic Compression," https://arxiv.org/abs/2408.17063

EXPOSITORY ARTICLES

[E02] Ali Atiia, K. Lee, "Sharded PIR Design for the Ethereum State," In *ethresearch*, 2026, https://ethresear.ch/t/sharded-pir-design-for-the-ethereum-state/24552

[E01] K. Lee, "The Dawn of Modern Cryptography (Series in Korean)," In *HORIZON*, 2025–2026, https://horizon.kias.re.kr/30970/, https://horizon.kias.re.kr/31730/, https://horizon.kias.re.kr/32364/, https://horizon.kias.re.kr/33039/

HONORS & AWARDS

- Distinguished Reviewer Award, S&P 2026 — May 2026
- Sejong Science Fellowship — 2024–2025
  National Research Foundation of Korea
  ≈$50,000/year
- Doctoral Dissertation Award — Apr 2024
  Korean Mathematical Society
  - Best Award ($1,000)
    "A Study on Homomorphic Packing: Definitions, Constructions, and Limitations"
- Doctoral Dissertation Award — Aug 2023
  College of Natural Sciences, Seoul National University
  - Best Award ($2,000)
    "A Study on Homomorphic Packing: Definitions, Constructions, and Limitations"
- Global PhD Fellowship — 2018–2023
  National Research Foundation of Korea
  Full Tuition and ≈$15,000/year
  - Award for Top 10% of Global PhD Fellowship ($4,000) — May 2022
  - Award for Top 10% of Global PhD Fellowship ($4,000) — Mar 2020
- National Cryptography Contest
  National Security Research Institute
  - Best Award ($3,000) — Oct 2022
    "Bit Security as Cost to Observe Advantage"
  - Best Award ($3,000) — Oct 2021
    "Limits of Polynomial Packings for $\mathbb{Z}_{p^k}$ and $\mathbb{F}_{p^k}$"
  - Excellence Award ($2,000) — Oct 2020
    "MHz2k: MPC from HE over $\mathbb{Z}_{2^k}$"
  - Excellence Award ($1,500) — Nov 2017
    Problem-solving Track
  - Excellence Award ($1,500) — Nov 2016
    "Privacy-Preserving Computation of Predictive Medical Models with Minimax Approximation"
- Best Paper Runner-up, Asiacrypt 2019 — Dec 2019
  International Association for Cryptologic Research
  "Numerical Methods for Comparison on Homomorphically Encrypted Numbers"
  Invited to *Journal of Cryptology* (Top 3 of 71 accepted papers among 307 submissions)

- First Place Prize, iDASH Genomic Data Privacy and Security Protection Competition — Oct 2017
  Track 3: Homomorphic Encryption (HME) based Logistic Regression Model Learning

PATENTS

[P05] Jung Hee Cheon, K. Lee, "Electronic Device And Server Device Responding To Encrypted Query And Methods Thereof," *KOR 1020240127246, US 18925390*

[P04] Jung Hee Cheon, Junyoung Jung, Taeseong Kim, K. Lee, Jai Hyun Park, "Electronic Apparatus For Bootstrap Processing Homomorphic Encrypted Messages And Methods Thereof," *KOR 1020240032226, US 12603754 granted*

[P03] Jung Hee Cheon, K. Lee, Sumin Kim, Jae Seon Kim, "Mobile Device For Detecting Route Overlap And Methods Thereof," *KOR 1026773130000 granted, US 11937078 granted*

[P02] Jung Hee Cheon, K. Lee, Dongwoo Kim, "Ciphertext Processing Method For Zero-Knowledge Proof And Apparatus Thereof," *KOR 1025994060000 granted*

[P01] Jung Hee Cheon, K. Lee, Dongwoo Kim, "Tweaked Interpolation For Multiparty Computation," *KOR 1022577790000 granted*

FUNDED PROJECTS

[F02] "Homomorphic Encryption and Its Application to Secure Multi-Party Computation" 2024–2025
Sponsor: National Research Foundation of Korea (Sejong Science Fellowship)
Total Amount: ≈$50,000
PI: K. Lee

[F01] "Lattice-based Fully Homomorphic Encryption and Its Application to Privacy-preserving Machine Learning" 2018–2023
Sponsor: National Research Foundation of Korea (Global PhD Fellowship)
Total Amount: ≈$100,000
PI: K. Lee

INVITED TALKS

- Oblivious Compression for Homomorphic Encryption and Its Applications — Feb 2025
  EIMS-KMS International Workshop on Cryptography, Virtual
- Homomorphic Encryption: An Introduction to Secure Computation — Nov 2024
  ESL GCI Seminar Series @ Rochester Institute of Technology, New York, USA
- On the Bit Security of Cryptographic Primitives — Oct 2022
  2022 Korean Mathematical Society International Conference, Seoul, Korea
  Invited Speaker of Focus Session on "Discrete Mathematics and Mathematics of Computer Science"
- Introduction to Secure Computation — Mar 2022
  BK21 Colloquium (Rookies Pitch) @ Seoul National University, Seoul, Korea
  Invited as an Outstanding Graduate Student of Math@SNU

PRESENTATIONS

- Barely Doubly-Efficient Private Information Retrieval — Nov 2026
  UMD Crypto Seminar
- Private Information Retrieval for Ethereum — Nov 2026
  Devcon, Mumbai, India
- Barely Doubly-Efficient SimplePIR
  Crypto 2026, Santa Barbara, CA — Aug 2026
  SNU Crypto Seminar — Aug 2026
- Privacy & Scaling Explorations at Ethereum Foundation — May 2025
  Silicon Valley Private AI Forum (K-PAI), Santa Clara, CA
- Oblivious Message Retrieval for ZCash — Jul 2024
  MPC & FHE Primer @ EDCON2024, Tokyo, Japan
- VeriSimplePIR: Verifiability in SimplePIR at No Online Cost for Honest Servers — Apr 2024
  Bay Area Crypto Day
- Limits of Polynomial Packings for $\mathbb{Z}_{p^k}$ and $\mathbb{F}_{p^k}$
  Eurocrypt 2022, Trondheim, Norway — May 2022
  2022 Korean Mathematical Society Spring Meeting — Apr 2022
- MHz2k: MPC from HE over $\mathbb{Z}_{2^k}$ with New Packing, Simpler Reshare, and Better ZKP
  Crypto 2021, Virtual — Aug 2021
  2020 Korean Mathematical Society Fall Meeting — Oct 2020

- Microsoft Private AI Bootcamp — Jul 2020
  2020 Korean Mathematical Society Spring Meeting
- Numerical Methods for Comparison on Homomorphically Encrypted Numbers — Apr 2019
  2019 Korean Mathematical Society Spring Meeting
- Privacy-preserving Predictive Models with Minimax Approx. and Non-adjacent Form — Apr 2017
  WAHC 2017, Sliema, Malta

EXPERIENCES

- Visiting Student (Prof. Vinod Vaikuntanathan) — Oct 2022–Dec 2022
  MIT, Boston, Massachusetts, USA
- Private AI Bootcamp — Dec 2019
  Team Project: *Ensuring Trust when Trading ML Models*
  Microsoft Research, Redmond, Washington, USA

SERVICES

- Program Committee
  Asiacrypt (2025, 2026), CCS (2026), S&P (2026), USENIX Security (2027), ISC (2025)
  - S&P 2026 Distinguished Reviewer Award
- Editorial Board
  IACR CiC (2025)
- Reviewer (Conferences)
  Crypto (2024, 2025, 2026), Eurocrypt (2025, 2026, 2027), Asiacrypt (2019, 2021, 2022, 2023, 2024), TCC (2024), PKC (2019, 2027), AsiaCCS (2023), CT-RSA (2019, 2020), PQCrypto (2020, 2023, 2024), ANTS (2020), FHE.org Workshop (2022), Mathcrypt Workshop (2023)
- Reviewer (Journals)
  Journal of Cryptology (JoC), Transactions on Dependable and Secure Computing (TDSC), Transactions on Information Theory (TIT), Designs, Codes and Cryptography (DCC)

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